

INDIAN KNOWLEDGE SYSTEM

Short Notes for Examination

Modules 1 – 5

MODULE 01	Society, State and Polity in India
MODULE 02	Indian Literature, Culture, Tradition
MODULE 03	Indian Religion, Philosophy and Practices
MODULE 04	Science, Management & Indian Knowledge System
MODULE 05	Cultural Heritage and Performing Arts

How to Use These Notes

- Focus on BOLD terms — likely exam MCQs
- REMEMBER boxes = high-yield facts
- Notes in purple = examiner's favourites
- ★ Tables = quick comparison — read twice

MODULE 1: SOCIETY, STATE AND POLITY IN INDIA

1.1 STATE IN ANCIENT INDIA — ORIGIN THEORIES

Evolutionary Theory	State evolved naturally: Family → Clan → Tribe → Kingdom. Human beings are inherently social. Key thinker: Aristotle — 'Man is a political animal'. In India — Mahajanapadas evolved from tribal confederacies.
Force Theory	State formed by conquest & coercion. Strong conquered the weak. Max Weber — 'State holds monopoly of legitimate force'. Examples: Nanda Empire, early Mauryan expansion.
Mystical / Divine Theory	King is divine or god's representative on Earth. Manusmriti: King created from particles of 8 guardian deities (Indra, Vayu, Yama, Sun, Fire, Varuna, Moon, Kubera). Concept of Chakravartin (universal ruler) & Devaraja (divine king).
Contract Theory	State formed by voluntary agreement. Indian version — Mahasammata ('the great elected one'). Buddhist text Digha Nikaya: People elected a king to maintain order; king gets 1/6 of produce in return. Arthashastra: Mutual obligation — king protects, subjects pay.

■ *Divine kingship ≠ Absolute kingship — King still bound by Dharma & Shastras.*

1.2 STAGES OF STATE FORMATION

- Janapada → Mahajanapada → Samrajya (Empire)
- **16 Mahajanapadas (600–300 BCE)**: Magadha, Kosala, Kasi, Vatsa, Anga, Avanti, Kamboja, Gandhara, Vriji...
- Magadha became most powerful under **Nandas** then **Mauryas**
- Transition from tribal to monarchical to imperial governance
- Archaeological evidence — Painted Grey Ware (PGW) & Iron Age linked to Janapada era

1.3 KINGSHIP

Rajadharma	King's primary duty — protection of subjects (Prajā)
Dandaniti	Science of governance & punishment — the rod of justice
4 Tools (Upaya)	Sama (conciliation) Dana (gift/bribe) Bheda (divide) Danda (force)
Rajyabhisheka	Sacred coronation ritual — legitimizes rule; priest + nobles involved
Chakravartin	Universal ruler — sovereign over all directions; ideal of Indian kingship
Constraints	King bound by Dharma, Shastra, Mantri Parishad — NOT absolute ruler

1.4 COUNCIL OF MINISTERS & ADMINISTRATION

■ Mantri Parishad (Arthashastra by Kautilya)

Mantrin / Amatya	Prime Minister / Chief Minister / Civil Servant
Purohita	Royal Priest — religious advisor; performed rituals

Senapati	Commander-in-Chief of armed forces
Yuvaraj	Crown Prince — heir to throne
Dauvarika	Chief of Palace Guards
Antapala	Governor of border territories (frontier regions)
Samaharta	Chief Revenue Collector
Sannidhata	Treasurer / Keeper of the stores

■ Administrative Divisions

- Empire → **Janapada** → **Sthaniya** → **Dronamukha** → **Kharvatika** → **Grama** (Village)
- Village headman = **Gramika**; District = **Ahara/Vishaya**
- Spies (**Gudhapurushas**) — Kautilya's elaborate espionage network
- Arthashastra describes **18 departments** of government

1.5 POLITICAL IDEALS — WELFARE & SAPTANGA

■ Welfare State Conditions (Arthashastra)

- *'In the happiness of his subjects lies the king's happiness; in their welfare, his welfare'* — **Kautilya**
- King's 4 duties: Protection from enemies | Internal law & order | Economic welfare | Spiritual guidance
- Famine relief, irrigation works, public roads — state responsibility
- 4 Ideals for governance: **Dharma, Artha, Kama, Moksha**

■ Saptanga — Seven Limbs of the State

1. Swami	The King — sovereign; fountain of authority
2. Amatya	Ministers & civil servants — administrative machinery
3. Janapada	Territory with population — resource base
4. Durga	Fortified capital city — seat of power & defense
5. Kosha	Treasury — economic foundation of the state
6. Danda	Army — coercive power; enforcement mechanism
7. Mitra	Allies — external support; foreign relations

■ **REMEMBER:** All 7 must function in harmony. Weak treasury (Kosha) = weak army = weak state.

1.6 SOCIETY IN ANCIENT INDIA

■ Purushartha — Four Aims of Life

Dharma	Moral/religious duty — the supreme guide for all actions; basis of social order
Artha	Material prosperity, wealth, political power — needed for sustaining life
Kama	Desire, love, aesthetic pleasures — governed by Dharma
Moksha	Liberation from cycle of rebirth (Samsara) — ultimate goal of human existence

■ *Dharma is the means; Moksha is the end. Artha & Kama must be pursued within Dharma.*

■ Varna System (Varṇashrama)

Brahmin	Priests, teachers, scholars — intellectual class; head of cosmic Purusha
Kshatriya	Warriors, rulers, administrators — protective class; arms of Purusha
Vaishya	Traders, farmers, artisans, moneylenders — economic class; thighs
Shudra	Service providers, labourers — servant class; feet of Purusha

- **Rigveda Purusasukta (X.90)** — metaphorical cosmic origin of 4 varnas from Purusha (cosmic man)
- **Manusmriti** — codified varna as birth-based & hereditary; legal disabilities for lower varnas
- **Jati** (sub-castes) developed later — more rigid; 3000+ jatis; occupational groups
- **Varna ≠ Jati**: Varna = 4 broad classes; Jati = thousands of sub-castes
- Buddhism & Jainism rejected birth-based varna; advocated social mobility through karma

■ Ashrama — Four Stages of Life

Brahmacharya (0–25 yrs)	Student life; celibacy; learning under Guru in Gurukul; acquisition of knowledge
Grihastha (25–50 yrs)	Householder stage; marriage, family, career; most important ashrama
Vanaprastha (50–75 yrs)	Forest dweller; gradual withdrawal from worldly life; contemplation
Sannyasa (75+ yrs)	Complete renunciation; wandering ascetic; focused on Moksha

■ *Ideal framework — not rigidly followed by all; mainly applicable to twice-born (dvija) varnas.*

■ Marriage in Ancient India — 8 Forms (Manusmriti)

1. Brahma	Best	Father gifts daughter to a learned man — gift with honor
2. Daiva	Good	Daughter given to officiating priest as sacrificial fee
3. Arsha	Approved	Groom gives a cow & bull; symbolic bride price
4. Prajapatya	Approved	Mutual consent; father gives daughter after blessing
5. Gandharva	Mixed	Love marriage; mutual consent; Shakuntala-Dushyanta example
6. Asura	Disapproved	Bride purchased with gifts & money

7. Rakshasa	Condemned	Abduction of bride after defeating her family
8. Paishacha	Worst	Marriage by deceit or force — condemned

- **Svayamvara** — Bride chooses her own husband (Kshatriya custom); Draupadi, Sita examples
- **Niyoga** — Levirate practice; widow bears child with deceased husband's brother

■ Gender as a Social Category

- **Vedic age:** Women relatively free — participated in sabhas, philosophical debates, composed hymns
- **Women Scholars:** Gargi (debated Yajnavalkya), Maitreyi (philosophical partner), Lopamudra (composed hymns)
- **Post-Vedic decline:** Increased restrictions — Manusmriti imposed severe limitations
- Manusmriti: Woman must be controlled by father (childhood) → husband → son (old age)
- **Devadasi system** — temple dancers; began as respected, later exploited

■ Challenges Faced by Women

Sati	Widow self-immolation on husband's pyre; increased in medieval period; banned 1829 by Ram Mohan Roy
Child Marriage	Common in medieval period; bride as young as 6–8 years; condemned by reformers
Purdah	Seclusion of women from public life; more prominent post-Muslim rule
Property Rights	Women denied inheritance; dowry (Stridhan) was only women's property
Devadasi	Temple women; originally revered; gradually exploited sexually
Female Infanticide	Son preference; girl babies killed; condemned in modern law

■ Four-Class Classification (Buddhist View)

- Buddhism **rejected** varna based on birth — class based on **occupation & conduct**
- Digha Nikaya: 4 classes = Khattiya (rulers), Brahmin (priests), Vessa (traders), Sudda (servants)
- Key principle: **Social mobility possible** through effort, virtue & right action
- Jataka tales show characters changing class through their actions

■ Slavery in Ancient India

- **Arthashastra (Kautilya)** — regulated slavery; 9 types listed
- Types: Debt-slaves, war captives, born slaves, self-sold, pledged, gift, inherited, penal, temple
- Protections: Master could not kill slave; slave could earn freedom by paying ransom
- **Buddhism & Jainism** — opposed cruel treatment; advocated compassion toward all
- Less rigid than Greek/Roman slavery — no mass plantation slavery in India

MODULE 2: INDIAN LITERATURE, CULTURE, TRADITION & PRACTICES

2.1 EVOLUTION OF SCRIPTS & LANGUAGES

Harappan Script (~2600–1900 BCE)	UNDECIPHERED till date ~400 signs identified Pictographic/logographic Right-to-left direction Found on seals, tablets, copper plates No bilingual text found (no Rosetta Stone)
Brahmi Script (~3rd c. BCE+)	Mother of most Indian scripts Ashoka's edicts first major evidence Left-to-right abugida Evolved into: Devanagari, Tamil, Kannada, Telugu, Bengali, Gujarati scripts
Kharosthi Script	Right-to-left direction NW India (Afghanistan-Pakistan) Used by Ashoka in NW edicts Died out unlike Brahmi
Classical Sanskrit	Panini's Ashtadhyayi (~500 BCE) — 4000 grammatical rules Most precise grammar system in ancient world

■ *Harappan Script — One of archaeology's greatest unsolved mysteries. 400+ signs, no bilingual key found.*

2.2 VEDIC LITERATURE

■ Four Vedas — The Foundation

Rigveda	Oldest religious text of any Indo-European language 1028 hymns (suktas) 10 Mandalas Dedicated to nature deities (Indra, Agni, Varuna, Soma)
Samaveda	Musical version of Rigveda hymns Set to musical notation for singing during soma rituals Origin of Indian classical music
Yajurveda	Prose prayers & mantras for yajna (fire sacrifice) rituals Two versions: Shukla & Krishna
Atharvaveda	Spells, charms, healing hymns Folk traditions, magic, medicine Contains earliest Ayurvedic references

- **4 Parts of each Veda:** Samhita (hymns) → Brahmana (ritual explanations) → Aranyaka (forest texts) → Upanishad (philosophy)

■ Upanishads (Vedanta — End of Vedas)

- **108 Upanishads total; 12 Principal ones** (Brihadaranyaka, Chandogya, Katha, Isha, Mundaka, Mandukya...)
- Key concepts: **Brahman** = Universal Soul (absolute reality) | **Atman** = Individual Soul
- **Tat Tvam Asi** — 'That Thou Art' (Chandogya Up.) — Atman = Brahman
- **Aham Brahmasmi** — 'I am Brahman' (Brihadaranyaka Up.)
- Philosophical dialogues — moved away from rituals toward inner self-realization
- **Shankaracharya's Advaita** based entirely on Upanishads

2.3 EPIC LITERATURE

	Valmiki — 'Adikavi' (First Poet)	Vyasa (Ved Vyasa)
	~24,000 shlokas — 7 Kandas	~100,000 shlokas — World's longest epic — 18 Parvas + Harivamsa
	Ideal life, Dharma vs Adharma, Rama's exile & rescue of Sita	Kurukshetra War between Pandavas & Kauravas

	Uttara Kanda (questioned as later addition)	Bhagavad Gita (Ch. 6, Bhishma Parva) — 18 chapters
	Rajadharma, Maryada Purushottam (Ideal Man), Ideal wife, loyalty	Karma Yoga, Bhakti Yoga, Gyana Yoga, Nishkama Karma
	Bala Ayodhya Aranya Kishkindha Sundara Yuddha Uttara	Adi Sabha Vana Virata Udyoga Bhishma Drona Karna Shalya Sauptika Stri Shanti Anushasana Ashvamedhika Ashramavasika Mausala Mahaprasthanika Svargarohana

■ **REMEMBER: 'What is not in Mahabharata is nowhere' — it is an encyclopedia of Indian civilization.**

■ Puranas — Encyclopedic Mythology

- **18 Mahapuranas** — encyclopedic texts covering mythology, cosmology, genealogy, geography
- **Pancha Lakshana** (5 characteristics): Sarga (creation) | Pratisarga (dissolution) | Vamsha (genealogy) | Manvantara (cosmic cycles) | Vamshanucharita (kings' histories)
- Most important: **Bhagavata Purana** (devotion to Vishnu/Krishna) | Vishnu | Shiva | Markandeya | Brahma
- **Bhagavata Purana** — 12 books; story of Krishna; foundation of Bhakti movement

2.4 BUDDHIST & JAIN LITERATURE

■ Buddhist Literature — Pali Canon (Tipitaka)

Vinaya Pitaka	Rules for monastic community (Sangha) — 227 rules for monks
Sutta Pitaka	Discourses of Buddha — Dhammapada (423 verses, ethical essence), Jataka tales (547 stories of past lives)
Abhidhamma Pitaka	Higher philosophical analysis of mind & matter
Milindapanha	Dialogue between King Menander (Milinda) & monk Nagasena — philosophical masterpiece

■ Jain Literature

- **Agamas** — canonical texts; 12 Angas (main scriptures); language: Ardhamagadhi Prakrit (early), later Sanskrit
- **Acharanga Sutra** — oldest Jain text; rules for monk's conduct; life of Mahavira
- **Kalpasutra** (Bhadrabahu) — biography of Mahavira and 23 Tirthankaras; Jain monks' lives
- **Tattvartha Sutra** (Umasvati) — philosophical summary; accepted by both Digambara & Shvetambara

2.5 KAUTILYA'S ARTHASHASTRA

- Written ~4th century BCE; rediscovered in 1905 by R. Shama Shastri at Mysore
- **15 Books** — covers: statecraft, economics, military, law, diplomacy, agriculture, espionage
- **Mandala Theory** — concentric circle foreign policy: neighbor = enemy; neighbor's neighbor = friend
- **Saptanga Theory** — 7 elements of state (see Module 1)
- **Four Upayas:** Sama, Dana, Bheda, Danda — tools of statecraft
- Most comprehensive political-economic treatise of the ancient world — compared to Machiavelli's Prince

2.6 FAMOUS SANSKRIT AUTHORS

Kalidasa (Gupta Era)	Shakuntala Meghaduta Raghuvarsha Kumarasambhava — Greatest Sanskrit poet
Bharavi	Kiratarjuniya — Arjuna vs Shiva (disguised as Kirata/hunter)

Magha	Shishupalavadha — Krishna kills Shishupala; elaborate poetics
Dandin	Dasakumaracharita (prose) Kavyadarsha (poetics manual)
Banabhatta	Harshacharita (biography of King Harsha) Kadambari (romance novel)
Vishakhadatta	Mudrarakshasa — Political drama of Chandragupta's rise
Shudraka	Mricchakatika (The Clay Cart) — social drama; merchant hero
Bhasa	13 plays discovered; Svapnavasavadatta — earliest dramatist

2.7 REGIONAL LITERATURES

■ Sangam Literature (Tamil) — 300 BCE to 300 CE

- Earliest Tamil poetry; **8 Anthologies (Ettutogai) + 10 Idylls (Pattupattu)**
- **Tolkappiyam** — oldest Tamil grammar; 3 books covering grammar, prosody, poetics
- Themes: **Akam** (love/interior) vs **Puram** (war/exterior/public life)
- **5 Tinai (landscapes)**: Kurinchi (mountains/union), Mullai (forest/waiting), Marutam (fields/infidelity), Neytal (seashore/longing), Palai (desert/separation)
- Greatest epics: **Silappathikaram** (Ilango Adigal) | **Manimekalai** — called 5th & 6th Vedas of Tamil

Telugu	Nannayya (1st poet, ~1000 CE) Tikkana Errana — 'Kavitrayam' (trio who translated Mahabharata). Allasani Peddana = 'Andhra Kavita Pitamaha'. Vijayanagara = Golden age.
Kannada	Pampa ('Adikavi Pampa', 941 CE) Ponna Ranna — 'Ratnatraya' (3 gems). Basavanna's Vachana literature (Lingayat movement). Pampa's Vikramarjuna Vijaya = 1st great Kannada work.
Malayalam	Developed from Tamil-Sanskrit blend (Manipravalam). Thunchathu Ramanujan Ezhuthachan = 'Father of Malayalam Literature'. Adhyatma Ramayanam his masterwork. Cherusseri — Krishnagatha.
Sangam	Earliest Tamil poetry (300 BCE–300 CE). Tolkappiyam = oldest grammar. Silappathikaram, Manimekalai = greatest epics.
Hindi	Tulsidas — Ramcharitmanas (Awadhi Hindi); Kabir — Dohas (couplets, Nirguna Bhakti); Surdas — Sursagar (Krishna devotion); Mirabai — devotional songs.
Persian	Amir Khusrau — 'Parrot of India'; Laila-Majnun, Khamsah. Mughal chronicles: Baburnama, Akbarnama, Ain-i-Akbari (Abul Fazl).
Urdu	Developed under Delhi Sultanate & Mughals. Mir Taqi Mir, Mirza Ghalib (Diwan = Urdu ghazal peak). Iqbal — 'Bang-e-Dara', 'Tarana-e-Hindi'.

MODULE 3: INDIAN RELIGION, PHILOSOPHY & PRACTICES

3.1 PRE-VEDIC RELIGION (HARAPPAN)

- **Mother Goddess** worship — terracotta female figurines found abundantly
- **Pashupati figure** (Proto-Shiva) — seal showing yogic figure surrounded by animals
- **Ritual bathing** — Great Bath at Mohenjo-daro; purification concept
- **Tree worship** (Pipal tree), **animal worship** (unicorn, humped bull on seals)
- **Fire altars** at Kalibangan — possible evidence of Vedic-type fire rituals
- No large temples found; domestic shrines suggest personal/household worship

3.2 VEDIC RELIGION

- **Nature worship** — Indra (thunder/rain), Varuna (cosmic order/water), Agni (fire), Soma (ritual drink)
- **Yajna (sacrifice)** — central ritual; Ashvamedha (horse), Rajasuya (royal consecration), Soma sacrifice
- **Rita** — cosmic moral order (precursor of Dharma); maintained by Varuna
- **Later Vedic:** Rise of Brahma-Vishnu-Shiva trinity; elaborate rituals; priestly dominance
- **Upanishadic period:** Shift from external ritual to inner realization; Brahman = ultimate reality

■ *Vedic religion = earliest layer of Hinduism. It evolved over 1500+ years.*

3.3 BUDDHISM

Four Noble Truths	1. Dukkha (life = suffering) 2. Samudaya (desire = cause) 3. Nirodha (cessation possible) 4. Magga (Eightfold Path = the way)
Eightfold Path	Right View Intention Speech Action Livelihood Effort Mindfulness Concentration
Three Jewels	Buddha (teacher) Dhamma (teaching) Sangha (community)
Three Marks	Anicca (impermanence) Dukkha (suffering) Anatta (no permanent self)
Nirvana	Liberation from cycle of rebirth (Samsara); extinguishing of desires
Pratityasamutpada	Dependent origination — everything arises due to conditions; nothing is independent

■ Buddhist Councils & Schools

Council	Location	Patron	Outcome
1st (483 BCE)	Rajagriha	Ajatashatru	Compiled Vinaya & Sutta Pitakas
2nd (383 BCE)	Vaishali	Kalashoka	Dispute on rules; first schism signs
3rd (250 BCE)	Pataliputra	Ashoka	Compiled Abhidhamma; missionary work
4th (1st c. CE)	Kashmir	Kanishka	Mahayana codified; split from Hinayana

- **Hinayana (Theravada)** — Older school; individual salvation; strict monastic code; prevalent in Sri Lanka, SE Asia
- **Mahayana** — Greater vehicle; Bodhisattva ideal (help all beings); prevalent in China, Japan, Korea, Tibet
- **Vajrayana** — Tantric Buddhism; secret rituals; prevalent in Tibet, Nepal, Bhutan

3.4 JAINISM

- Founded by **Vardhamana Mahavira** — 24th Tirthankara; attained Kevala Jnana (omniscience) at 42
- Earlier Tirthankaras: **Rishabhanatha** (1st; mentioned in Rigveda), **Parshvanatha** (23rd; 877–777 BCE)
- Jainism rejects Vedic authority but believes in karma & reincarnation

■ Five Great Vows (Pancha Mahavratas)

Ahimsa	Non-violence (supreme virtue) — not harming any living being in thought, word, deed
Satya	Truthfulness — speak only beneficial truth
Asteya	Non-stealing — not taking anything not given
Brahmacharya	Celibacy — added by Mahavira; Parshvanatha had only 4 vows
Aparigraha	Non-possessiveness — limit attachments & possessions

- **Digambara vs Shvetambara** — split ~1st century CE; Digambara monks wear no clothes (sky-clad); Shvetambara wear white
- **Anekantavada** — doctrine of many-sidedness: truth has multiple aspects/perspectives
- **Syadvada** — 'maybe/perhaps' — 7-fold conditional predication (Saptabhangi)
- **Jiva** (soul) vs **Ajiva** (non-soul) — 2 fundamental categories of Jain metaphysics

3.5 SIX SYSTEMS OF PHILOSOPHY (SHAD DARSHANAS)

Nyaya	Gautama Logical analysis; 4 valid sources of knowledge (Pramanas); syllogistic reasoning
Vaisheshika	Kanada Atomism — Paramanu (atom) as smallest unit; 6 categories (Padarthas) of reality
Samkhya	Kapila Dualism — Purusha (consciousness, inactive) + Prakriti (matter, active); 25 tattvas
Yoga	Patanjali Practical path; Ashtanga Yoga — Yama, Niyama, Asana, Pranayama, Pratyahara, Dharana, Dhyana, Samadhi
Mimamsa	Jaimini Vedic ritual interpretation; karma & sacrifice lead to liberation; Dharma = Vedic injunctions
Vedanta	Badarayana (Brahma Sutras) Brahman as ultimate reality; Upanishads interpreted. Advaita (Shankara), Vishishtadvaita (Ramanuja), Dvaita (Madhva)

■ **REMEMBER:** All 6 are **ASTIKA** (accept Vedas as authority). **Buddhism, Jainism, Charvaka = NASTIKA** (reject Vedas).

3.6 SHANKARACHARYA & PHILOSOPHICAL DOCTRINES

Advaita (Shankara, 788–820 CE)	Non-dualism Brahman alone real; world = Maya (illusion) Jiva = Brahman 4 Mathas: Sringeri (S), Puri (E), Dwaraka (W), Jyotirmath/Badrinath (N)
Vishishtadvaita (Ramanuja)	Qualified Non-dualism Individual souls & world real but exist within Brahman Bhakti = path to liberation Sri Vaishnavism
Dvaita (Madhvacharya)	Dualism Vishnu & souls permanently distinct Vishnu is supreme Tattvavada philosophy
Achintya Bhedabheda (Chaitanya)	Inconceivable difference & non-difference Soul both same & different from Brahman Gaudiya Vaishnavism

3.7 HETERODOX SECTS (NASTIKA)

- **Charvaka/Lokayata** — Pure materialism; only 4 elements; no soul, no afterlife; 'Eat, drink, be merry'; reject Vedas
- **Ajivika** — Founded by Makkhali Gosala; strict determinism (Niyati); no free will; all souls will be liberated after fixed cosmic cycles
- **Buddhism** — Rejects Vedic authority & caste; accepts karma & rebirth; Middle Path
- **Jainism** — Rejects Vedas; accepts karma; extreme non-violence; universe is eternal

3.8 BHAKTI & SUFI MOVEMENTS

■ Bhakti Movement — Key Saints

Region	Saint(s)	Contribution
Tamil (S. India)	12 Alvars 63 Nayanmars	Alvars: Divya Prabandham (4000 hymns) — Vaishnavism. Nayanmars: Tevaram — Shaivism.
Karnataka	Basavanna (12th c.)	Vachana literature; Lingayat movement; rejected caste & rituals
Maharashtra	Tukaram, Eknath, Namdev, Ramdas	Abhanga poetry; Vitthal/Vithoba devotion; Varkari tradition
North India	Ramananda, Kabir, Mirabai, Surdas, Tulsidas	Ramananda: accepted all castes. Kabir: Hindu-Muslim unity, Dohas. Mirabai: Krishna devotion, defied social norms.
Bengal	Chaitanya Mahaprabhu (15th c.)	Gaudiya Vaishnavism; ecstatic devotion (Kirtana); Hare Krishna tradition
Punjab	Guru Nanak (1469–1539)	Founded Sikhism; monotheism; equality; Gurmukhi script

■ *Bhakti challenged caste hierarchy, ritualism, and Brahmin dominance — democratized religion.*

■ Sufi Movement

- Islamic mysticism; introduced in India ~10th century CE
- **Chishti Order** (most popular in India) — Moinuddin Chishti (Ajmer); Nizamuddin Auliya (Delhi)
- Other orders: **Suhrawardi, Qadiri, Naqshbandi**
- **Silsila** — spiritual chain of succession (pir to mureed)
- **Sama (Qawwali)** — ecstatic devotional music; Amir Khusrau composed earliest Qawwalis
- Hindu-Muslim syncretism — bridged communities; concept of unity in God (Tawhid)

3.9 SOCIO-RELIGIOUS REFORM MOVEMENTS (19th CENTURY)

Brahmo Samaj (1828)	Ram Mohan Roy Monotheism; against Sati, idol worship, caste Rationalist Hinduism Merged with other groups as Adi Brahmo Samaj
Arya Samaj (1875)	Dayananda Saraswati 'Back to Vedas'; rejected idols, caste, child marriage Shuddhi (reconversion) Founded DAV schools
Ramakrishna Mission (1897)	Vivekananda 'Service to man = service to God' Practical Vedanta Chicago Speech (1893) — Hindu renaissance
Prarthana Samaj (1867)	Ranade, Bhandarkar Maharashtra reform; widow remarriage, women's education
Theosophical Society (1875)	Blavatsky + Olcott Hindu/Buddhist revival; Annie Besant — Home Rule League

Singh Sabha (1873)	Punjab Sikh reform; challenged caste, superstition in Sikhism Gurumukhi literacy
Aligarh Movement (1875)	Sir Syed Ahmed Khan Muslim education; English medium; founded MAO College (now AMU)

MODULE 4: SCIENCE, MANAGEMENT & INDIAN KNOWLEDGE SYSTEM

4.1 ASTRONOMY IN INDIA

Vedanga Jyotisha (~1400 BCE)	Earliest astronomical text; determined time of Vedic rituals; 27 Nakshatras identified
Aryabhata (476–550 CE)	Aryabhata: Earth rotates on its own axis Real cause of eclipses $\pi \approx 3.1416$ Day = 23hr 56min Heliocentric hints
Varahamihira (505–587 CE)	Panchasiddhantika — 5 astronomical systems compiled Brihat Samhita — encyclopedia of astronomy
Brahmagupta (598–668 CE)	Brahmasphutasiddhanta Rules for zero & negatives Earth's gravity mentioned More accurate earth circumference
Bhaskaracharya (1114–1185 CE)	Siddhantashiromani Lilavati (mathematics) Bijaganita (algebra) Near-calculus concepts
Jantar Mantar (18th c.)	Observatories built by Maharaja Jai Singh II in Delhi, Jaipur, Varanasi, Ujjain, Mathura

4.2 MATHEMATICS IN INDIA

Zero & Decimal System	India gave world place-value decimal notation. Brahmagupta (628 CE) defined rules for zero arithmetic. Aryabhata used zero as placeholder.
Pi (π)	Aryabhata: $\pi = 3.1416$ Madhava (Kerala school, 14th c.): π to 11 decimal places using infinite series
Trigonometry	Aryabhata introduced Jya (sine), Kojya (cosine), Utkrama-jya (versine) — revolutionized spherical astronomy
Algebra	Brahmagupta: solved quadratic equations Bhaskara: solved Pell's equation using Chakravala method
Fibonacci Numbers	Pingala's Chandah-shastra (~300 BCE) contains Fibonacci sequence in study of Vedic meter (Matra Meru)
Kerala School	Madhava, Nilakantha — infinite series for sine, cosine, arctangent — 200 years before Newton/Leibniz
Sulba Sutras	~800–500 BCE Geometric constructions; Pythagorean theorem stated; irrational numbers

■ **REMEMBER:** India invented zero → decimal system → modern mathematics → computing. Greatest intellectual contribution.

4.3 MEDICINE IN INDIA — AYURVEDA

- **Ashtanga Ayurveda** — 8 branches: Internal medicine | Surgery | ENT | Toxicology | Psychiatry | Pediatrics | Rejuvenation | Aphrodisiac
- **Tridosha Theory:** Vata (air/space) + Pitta (fire/water) + Kapha (water/earth) — balance = health
- **Panchakarma** — 5 purification therapies: Vamana, Virechana, Basti, Nasya, Raktamoksha

Charaka Samhita (~2nd c. BCE)	Internal medicine (Kayachikitsa) 8 books, 120 chapters First systematic classification of diseases Diet & drug therapy
Sushruta Samhita (~3rd c. BCE)	'Father of Surgery' 120+ surgical instruments described Rhinoplasty, cataract surgery, caesarean section 300 surgical procedures 8 types of surgery

Ashtanga Hridayam
(Vagbhata, ~7th c.)

Concise encyclopedia of Ayurveda | Synthesizes Charaka & Sushruta |
Most used textbook in Kerala Ayurveda

■ *Sushruta performed plastic surgery 2,500 years before Europeans. Rhinoplasty still called 'Indian method' in West.*

4.4 METALLURGY IN INDIA

Iron Age	Iron smelting from ~1200 BCE Early Iron Age coincides with later Vedic period PGW (Painted Grey Ware) culture
Iron Pillar of Delhi	Gupta era (~375 CE) 7.2 m tall 99.72% pure iron No rust after 1600+ years Phosphorus-rich protective layer — unknown to modern science
Wootz Steel	'Damascus Steel' — originated in India High-carbon crucible steel Exported to Middle East, Rome Distinctive wavy pattern
Zinc Smelting	Zawar mines, Rajasthan — world's oldest zinc smelting site (9th century CE) 400 years before Europeans
Harappan Metals	Bronze (tin+copper alloy) from 2500 BCE Copper tools, bronze Dancing Girl Gold & silver jewelry with granulation technique

4.5 CHEMISTRY & PHYSICS IN INDIA

■ Chemistry (Rasashastra)

- **Rasashastra** — alchemy & iatrochemistry; medicinal use of metals & minerals
- **Nagarjuna** (~1st CE) — metallurgy & alchemy; Rasaratnakara text; preparation of mercury compounds
- **Rasa Ratna Samucchaya** — comprehensive chemistry text; preparation of drugs from metals
- Dyes (indigo), perfumes, soap, wax candles — well-developed industries
- **Arthashastra** — describes fire weapons (agni-yoga), sulphur-based incendiaries

■ Physics (Vaisheshika Atomism)

- **Kanada** (founder of Vaisheshika) — atomic theory: **Paramanu** = smallest indivisible particle
- 2 Paramanu → Dvyanuka (diatomic) → 3 Dvyanuka → Tryanuka (visible matter)
- **Gravity concept**: Varahamihira — 'Gurutva' (heaviness); objects fall due to earth's inherent property
- Theory of sound — Vaisheshika: sound travels through medium (akasha)

4.6 AGRICULTURE & WATER MANAGEMENT

■ Agriculture

- **Harappan agriculture** — wheat, barley, cotton; **first in world to grow cotton**; sesame, mustard
- **Krishi Parashara** — Sanskrit agricultural manual; soil types, seasons, crop selection, irrigation
- **Arthashastra** — superintendent of agriculture (Sitadhyaksha); crop rotation, irrigation taxes
- Spices trade — India dominated world spice trade (pepper, cardamom, turmeric, ginger)

■ Water Management

Harappan Drainage	World's earliest urban sanitation system (~2500 BCE); covered drains; soak pits; public wells
Sudarshana Lake	Mauryan irrigation dam near Girnar, Gujarat; Chandragupta period; repaired by Ashoka & later Rudradaman
Grand Anicut (Kallanai)	Karikala Chola (~2nd c. CE); dam on Kaveri River; still functional after 2000 years

Rani ki Vav	Stepwell at Patan, Gujarat (11th c.); UNESCO World Heritage; ornate multi-storey stepwell
Karez System	Underground channels in NW India; carries groundwater from hills to plains
Sringaverapura	Elaborate water harvesting system near Allahabad; Ramayana period (~1st c. BCE)

4.7 HARAPPAN TECHNOLOGIES

Town Planning	Grid layout; standardized bricks (ratio 1:2:4); covered drains; citadel + lower town; streets oriented N-S & E-W
Weights & Measures	Standardized weights (binary system 1,2,4,8,16,32...) and decimal system; accurate to 0.05mm
Pottery	Wheel-thrown; kiln-fired; red & black painted; standardized forms; found from Harappa to Gujarat
Cotton Textiles	World's FIRST cotton weavers (~2500 BCE); evidence of cotton threads & impressions on pottery
Dockyard	Lothal, Gujarat — world's earliest dockyard; tidal lock system; sea trade to Mesopotamia
Seals & Writing	Steatite seals; fired for hardness; boss on back for grip; used in trade/identification
Bronze Casting	Lost-wax (cire perdue) technique; Dancing Girl figure — remarkable naturalism

4.8 TRADE IN ANCIENT INDIA

- **Silk Route** — India connected Central Asia, Persia, Rome, China; silk, spices, gems traded
- **Indian Ocean trade** — monsoon winds used ('Hippalus' winds discovered by/named after Greek sailor)
- **Roman coins** found abundantly in South India — evidence of massive Rome-India trade
- **Shreni (Guilds)** — regulated trade & crafts; had legal status; acted like banks
- **India's GDP share** — ~25–30% of world GDP until 18th century (Angus Maddison data)
- Exports: Pepper, cotton, ivory, spices, gems, indigo | Imports: Gold, silver, wine, copper
- **Lothal** — major port town; trade with Mesopotamia (Dilmun/Bahrain) proven by shared seals

4.9 TEXTILE, WRITING & PYROTECHNICS

Textile Technology	Harappans = world's first cotton weavers Silk weaving in ancient India Muslin of Dhaka (Woven Air / Woven Wind) — so fine it was called 'running water' Block printing developed in Rajasthan Calico (from Calicut/Kozhikode) Kashmiri pashmina
Writing Technology	Harappan script (~2600 BCE) Brahmi script (3rd c. BCE) Kharosthi (NW) Grantha script (Sanskrit in South India) Devanagari evolved from Brahmi ~7th c. CE Birch bark & palm leaf manuscripts
Pyrotechnics	Arthashastra mentions agni-yoga (fire weapons) Sulphur, charcoal, saltpetre combinations Fireworks for celebrations mentioned in Mughal era Shipra festival fireworks Gunpowder knowledge predates Chinese claim in Indian texts

MODULE 5: CULTURAL HERITAGE AND PERFORMING ARTS

5.1 ARCHITECTURE IN ANCIENT INDIA

■ Temple Architecture — Three Major Styles

Feature	NAGARA (North)	DRAVIDA (South)	VESARA (Mixed)
Main Tower	Shikhara — curvilinear (beehive shape)	Vimana — stepped pyramidal tower	Blend of both styles
Entry Tower	No large gopuram	Large Gopurams at gates	Moderate size
Examples	Khajuraho, Konark Sun Temple, Somnath	Brihadeeswarar (Tanjore), Meenakshi	Hoysala temples (Belur, Halebid)
Region/Period	North & Central India; 6th–13th c.	Tamil Nadu, Kerala; 7th–16th c.	Karnataka; 12th–13th c.
Patron Dynasty	Gurjara-Pratiharas, Chandellas, Solankis	Pallavas, Cholas, Pandyas, Vijayanagara	Hoysalas, Chalukyas

■ Stupa Architecture

- **Stupa** = Buddhist burial/relic mound | Major: Sanchi (Ashoka), Amaravati, Sarnath, Bharhut
- **Parts:** Anda (dome) → Harmika (square railing on top) → Yashti (mast/umbrella) → Vedika (railing) → Torana (gateway) → Pradakshinapatha (circumambulation path)
- **Sanchi Stupa** — 3rd century BCE; Ashoka; 4 toranas with narrative carvings — masterpiece
- **Rock-cut architecture:** Ajanta (paintings, 2nd BCE–7th CE) | Ellora (30 caves, Buddhist+Hindu+Jain) | Kailasa Temple (Ellora) = largest monolithic rock-cut structure

5.2 SCULPTURES

Indus Valley	Terracotta figurines Dancing Girl (bronze, lost-wax, ~2500 BCE) Priest-King (steatite) Remarkable naturalism
Mauryan (3rd BCE)	Polished sandstone pillars Yaksha/Yakshi figures Lion Capital (Sarnath) = National Emblem Mauryan polish = mirror-like finish
Gandhara School (1st-5th c. CE)	Greco-Buddhist style Greek Apollo-like features on Buddha Wavy hair, toga-like robes NW India (modern Pakistan-Afghanistan)
Mathura School (1st-3rd c. CE)	Indigenous Indian style Thin muslin-clad figures Red Sikri sandstone First standalone Buddha images
Amaravati School	Deccan (Andhra); white limestone Narrative panels; dynamic figures Buddhist themes Influenced SE Asian art
Gupta Period (4th-6th c. CE)	Classical ideal Sarnath Buddha - transparent robe, spiritual serenity Perfect proportions Golden Age of Indian sculpture

5.3 COINS, SEALS & POTTERY

Punch-marked Coins (6th BCE)	Earliest Indian coins Symbols punched on silver/copper No king's image Janapada period
Indo-Greek Coins	Bilingual (Greek + Brahmi) King's portrait on obverse Most artistic ancient coins Evidence of Indo-Greek kings

Kushana Coins	Gold Dinars Kanishka's coins show Buddhist, Hindu, Zoroastrian gods — cultural fusion
Gupta Gold Dinars	Finest artistic achievement King shown in various activities Chandragupta II, Samudragupta
Medieval Coins	Sultans introduced Islamic calligraphy on coins; Akbar's coins had Ram-Sita images (early); later standardized

■ Pottery Sequence (Important for Chronology)

Pottery Type	Period	Associated Culture	Significance
Ochre Coloured Pottery (OCP)	2000–1500 BCE	Post-Harappan/Copper Hoard	Upper Gangetic plain
Painted Grey Ware (PGW)	1200–600 BCE	Later Vedic / Mahajanapadas	Grey with black geometric designs
Northern Black Polished (NBP)	700–200 BCE	Urban / Elite / Buddhist	Mirror-like polish; elite use; Buddhist sites
Red Ware	Throughout	Most common pottery	Widest distribution across all periods

5.4 CLASSICAL DANCE FORMS OF INDIA

Dance Form	State	Key Features & Origin
Bharatanatyam	Tamil Nadu	Oldest classical dance Devadasi temple tradition Abhinaya (expression) + Nritta (pure dance) Natya Shastra-based
Kathak	North India	Persian/Mughal influence Rapid footwork (tatkar) Fast spins (chakkar) Two gharanas: Jaipur & Lucknow
Odissi	Odisha	Lyrical, sculptural Tribhanga posture (3-bend) Derived from Mahari (temple dancers) Jayadeva's Gita Govinda themes
Manipuri	Manipur	Soft, gentle, no sharp angles Vaishnavite themes Sankirtana (group dance) No stamp on ground
Kuchipudi	Andhra Pradesh	Theatrical, dramatic Involves dialogue Brahmin village in Krishna district; originated from Bhagavata Mela
Mohiniyattam	Kerala	Feminine, lyrical White & gold costume Lasya style (graceful) Mohini (enchantress) theme
Kathakali	Kerala	Elaborate make-up & costume Colour codes for character (green=heroic) Night-long performances Male dancers
Sattriya	Assam	Established by Shankaradeva (15th c.) Vaishnavite Bhaonas (dramatic plays) Newest addition to classical list (2000)

■ *Natyashastra* by Bharata Muni — 'Panchamaveda' (5th Veda); foundational text for all performing arts; 36 chapters.

5.5 MUSIC IN INDIA

■ Two Classical Music Systems

Feature	HINDUSTANI (North)	CARNATIC (South)
Origin	Persian + indigenous; Mughal court influence	Pure Indian; based on Natyashastra

Form	Raga + Tala; improvisation-heavy (Alap)	Kriti (composition) — more structured; less improvisation
Ragas	Time-based & season-based ragas strictly followed	Not strictly time-based; more freedom
Instruments	Sitar, Sarod, Sarangi, Tabla, Bansuri	Veena, Mridangam, Violin, Ghatam
Trinity	Tansen (Akbar's court) — greatest maestro	Tyagaraja Muthuswami Dikshitar Syama Shastri
Key Ragas	Bhairav (dawn), Yaman (evening), Bhairavi (farewell)	Bhairavi, Kalyani, Todi, Shankarabharanam

5.6 THEATRE, DRAMA & NATYASHASTRA

- **Natyashastra** (Bharata Muni) — 36 chapters; covers drama, dance, music; 'Panchamaveda'
- **9 Rasas (Navarasa):** Shringara (love) | Hasya (humour) | Karuna (compassion) | Raudra (fury) | Vira (heroism) | Bhayanaka (terror) | Bibhatsa (disgust) | Adbhuta (wonder) | Shanta (peace) [added later]
- **Famous Sanskrit Dramas:** Shakuntala (Kalidasa) | Mricchakatika (Shudraka) | Mudrarakshasa (Vishakhadatta) | Svapnavasavadatta (Bhasa)
- **Kutiyattam** (Kerala) — oldest surviving Sanskrit theatre; UNESCO Intangible Cultural Heritage

Folk Theatre Forms

Form	Region	Theme
Yakshagana	Karnataka	Hindu myths; elaborate costume; all-night performance
Tamasha	Maharashtra	Social satire; comedy; Lavani songs
Nautanki	UP & Rajasthan	Romantic & historical stories; high-pitched singing
Jatra	Bengal & Odisha	Dramatic; loud dialogues; mythological stories
Therukoothu	Tamil Nadu	Street theatre; night performances; Mahabharata themes
Bhand Pather	Kashmir	Comedy & social commentary; mask-based
Ankiya Nat	Assam	Developed by Shankaradeva; Vaishnavite content

5.7 PAINTING IN INDIA

Bhimbetka Rock Art (~30,000 BCE)	Oldest cave paintings Madhya Pradesh Hunting, dancing, animals Pre-historic art spanning 10,000+ years
Ajanta Murals (2nd BCE–7th CE)	Tempera technique (not fresco) Buddhist themes 30 caves UNESCO World Heritage Masterpiece of ancient world art
Mughal Miniature (16th–18th c.)	Persian influence + Indian elements Portraits, court scenes, nature Akbar, Jahangir, Shah Jahan as patrons Jahangir — keenest art patron
Rajput Miniature	Bold colors Ramayana, Krishna themes Rajasthan & Pahari sub-schools Mewar, Marwar, Bundi styles
Pahari Miniature	Delicate; Kangra & Basohli schools Love themes (Radha-Krishna) Hill states of Himachal Pradesh
Kerala Murals	Large-scale temple paintings Mattancherry Palace, Padmanabhapuram Distinctive technique & vivid colors
Folk Art	Madhubani (Bihar-Mithila) Warli (Maharashtra) Pattachitra (Odisha) Gond (Madhya Pradesh) Kalamkari (Andhra)

5.8 PUPPETRY, MARTIAL ARTS & FAIRS

■ Puppetry (One of World's Oldest Art Forms)

String Puppets (Kathputli)	Rajasthan — most well-known; colourful costumes; elaborate stories
Shadow Puppets	Tholu Bommalata (AP) Ravana Chhaya (Odisha) Tolpava Koothu (Kerala) — leather puppets
Rod Puppets	Putul Nach (Bengal) Kundhei (Odisha) — rod-controlled
Glove Puppets	Pavakoothu (Kerala) Sakhi Kundhei (Odisha) — hand-operated

■ Martial Arts Traditions

Kalaripayattu	Kerala Oldest surviving martial art 'Mother of all martial arts' Influenced SE Asian martial arts 3 stages: Meithari, Kolthari, Angathari
Silambam	Tamil Nadu Stick fighting; also includes weapons training Mentioned in Sangam literature
Gatka	Punjab/Sikh Sword & stick; spiritual dimension Practiced during Gurmurb celebrations
Thang-ta	Manipur Sword (Thang) & spear (Ta); associated with Meitei culture
Malla Yuddha	Ancient wrestling; described in Ramayana (Vali-Sugriva), Mahabharata (Bhima-Jarasandha)

■ Major Fairs & Festivals

- **Kumbh Mela** — World's largest human gathering; 4 locations: Prayagraj, Haridwar, Ujjain, Nashik; UNESCO Intangible Heritage
- **Pushkar Fair** (Rajasthan) — Camel fair + pilgrimage to Brahma temple
- **Rath Yatra** (Puri, Odisha) — Chariot festival of Lord Jagannath; UNESCO recognized
- **Harvest Festivals:** Onam (Kerala) | Bihu (Assam) | Pongal (Tamil Nadu) | Baisakhi (Punjab)
- **Hemis Festival** (Ladakh) — Buddhist mask dance; celebrates Guru Padmasambhava
- **Losar** — Tibetan/Buddhist New Year; celebrated in Ladakh, Sikkim, Arunachal

5.9 INDIA'S CULTURAL CONTRIBUTION TO THE WORLD

Zero & Decimal System	Changed mathematics, science, and computing globally. Without zero, no binary code, no computers.
Chess (Chaturanga)	Invented in India (~6th century CE) Spread to Persia (Shatranj) → Arab world → Europe
Yoga & Ayurveda	Globally adopted wellness systems 21 June = International Yoga Day (proposed by India at UN)
Buddhism	Spread to East & Southeast Asia — cultural, philosophical, artistic unification
Cotton & Spices	India was primary supplier; drove world trade routes; Columbus sought India when he found America
Sanskrit & Scripts	Linguistic ancestor of SE Asian scripts; Khmer, Thai, Balinese derive from Indian scripts
Greater India	Cultural colonies: Angkor Wat (Cambodia) — largest Hindu temple; Borobudur (Indonesia) — largest Buddhist temple
Mathematics & Science	Trigonometry, algebra, astronomy — transmitted to Arab world → Europe (via Arab scholars)

Indigo & Cotton	India monopolized global textile trade; 'Indigo' from India (Indikón); Calico from Calicut
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5.10 INDIAN CINEMA

1913	Raja Harishchandra — Dadasaheb Phalke First Indian feature film Silent film Phalke = 'Father of Indian Cinema'
1931	Alam Ara (Ardeshir Irani) — First Indian talkie First sound film
1937	Kisan Kanya — First Indian colour film
Golden Age (1940–60s)	Satyajit Ray (Pather Panchali 1955), Guru Dutt, Bimal Roy, V. Shantaram — Indian cinema gets international recognition
Satyajit Ray	Apu Trilogy Academy Honorary Award Pather Panchali won at Cannes (1956)
Industries	Bollywood (Hindi/Mumbai) Tollywood (Telugu/Hyderabad) Kollywood (Tamil/Chennai) Mollywood (Malayalam/Kerala)
Scale	~1800+ films/year 20+ languages Largest film industry in world by NUMBER of films produced
Dadasaheb Phalke Award	India's highest film honour Given annually since 1969

■ ALL THE BEST FOR YOUR EXAMINATION! ■

Remember: Understand concepts first, then memorize key terms. Revise tables 2–3 times.